GATE-ALL BRANCHES

ENGINEERING MATHEMATICS

PROBABILITY AND STATISTICS

Lecture No.- 10



By- Rahul Sir

Topics covered in previous lecture



Topic

Probability density function

Topic

Question based on RV

Topics to be Covered



Topic — Probability density function

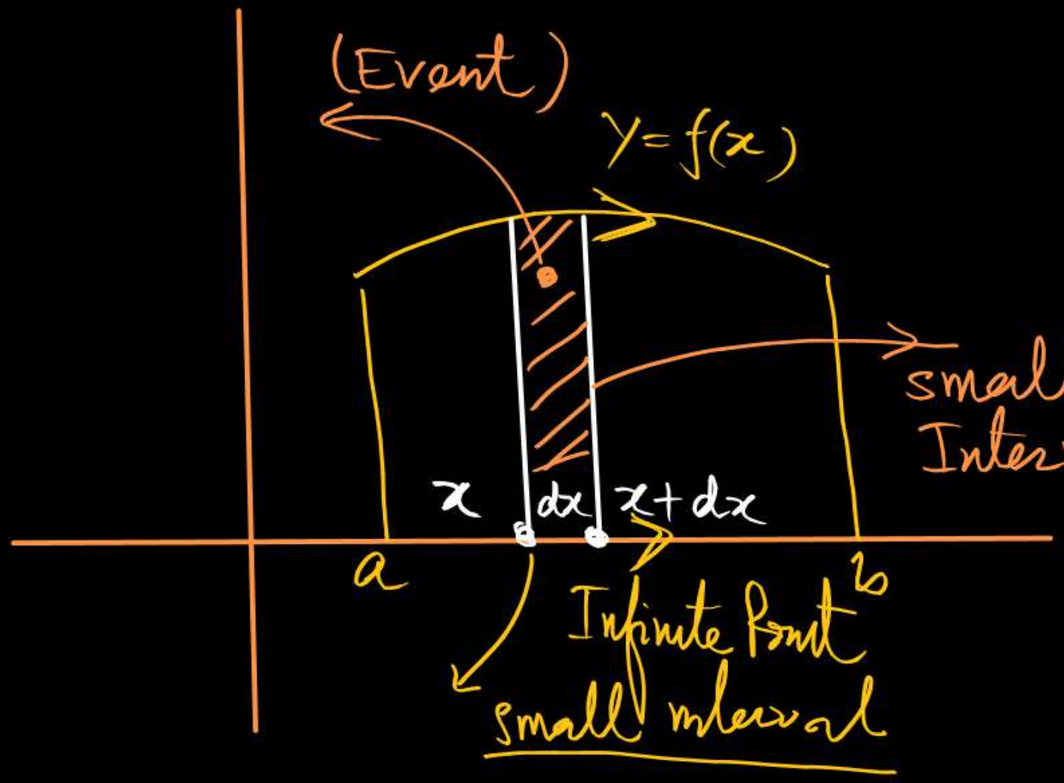
Topic — Question based on RV

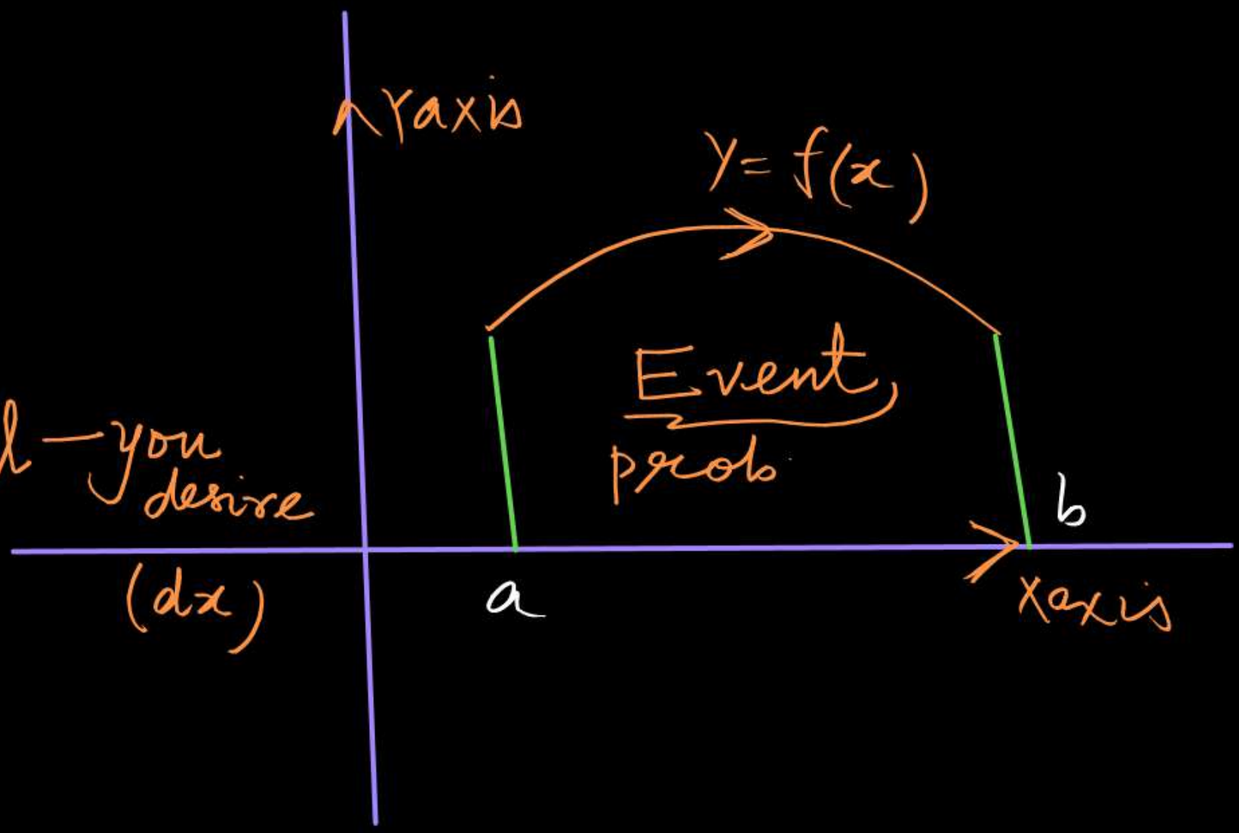
Topic — Expectation of a RV

Cumulative Distribution Function $\longrightarrow F_X(x_i) = P[X \le x_i]$

Probability density Function : Prob. Density Function Defined on continuous Random variable

$Y = f(x) \quad \forall \; x \in [a, b]$





Using cumulative Distribution Property = AREA

$$P(\underbrace{x}_{\text{Lower}} \le X \le \underbrace{x+dx}_{\text{upper}}) = F_X(x+dx) - F_X(x)$$

$F_X(x+dx) - F_X(x)$ = Area of this region $(x \le X \le x+dx)$

$$F_X(x+dx) - F_X(x) = \int f(x)\,dx = l \times b$$

$$\frac{F_X(x+dx) - F_X(x)}{dx} = f(x)$$

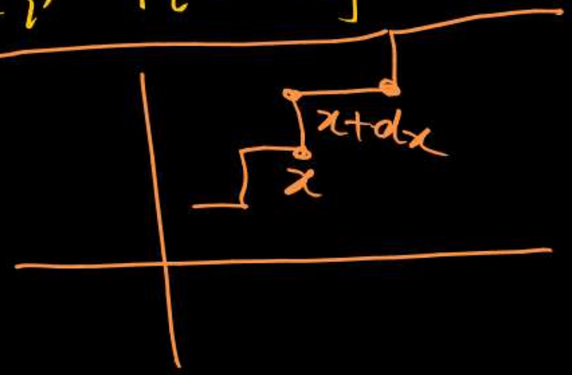
Area of small Interval $= l \times b = \int f(x)\,dx$



X = Continuous random variable

$[x \le X \le x+dx]$

$F_X(x_i) = P[X \le x_i]$



$$\frac{F_X(x+dx) - F_X(x)}{dx} = f(x)$$

$$\lim_{dx \to 0} \frac{F_X(x+dx) - F_X(x)}{dx} = f(x)$$

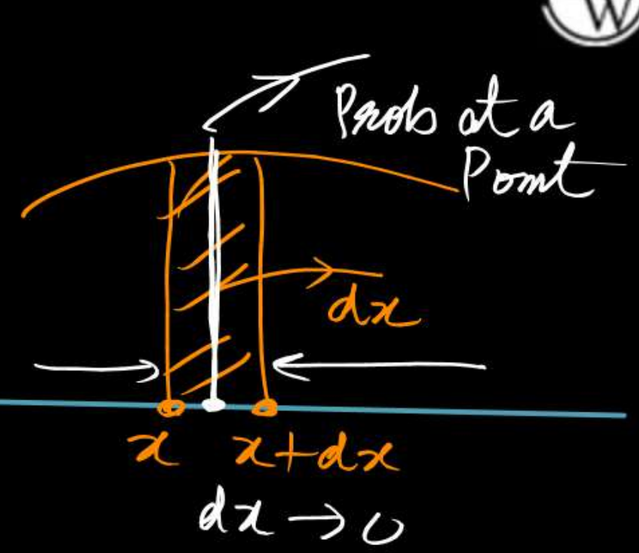
$$\boxed{F_X'(x) = f(x)}$$

both sides Integrate It

$$\int F_X'(x) = \int \frac{d}{dx}\left(F_X(x)\right) = \int f(x)\,dx$$

$$\boxed{F_X(x) = \int_{\text{Region}} f(x)\,dx}$$

cdf for continuous random variable.



$$F_X'(x) = \lim_{dx \to 0} \frac{F_X(x+dx) - F_X(x)}{dx}$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$F_X(x) = \int_{\text{Region}} f(x)\,dx$$

$f(x)$ = prob. density Function

If X is a continuous random variable $a \le X \le b$

$$F_X(b) - F_X(a) = \int_a^b f(x)\,dx$$

✓ $f(x)$ is given

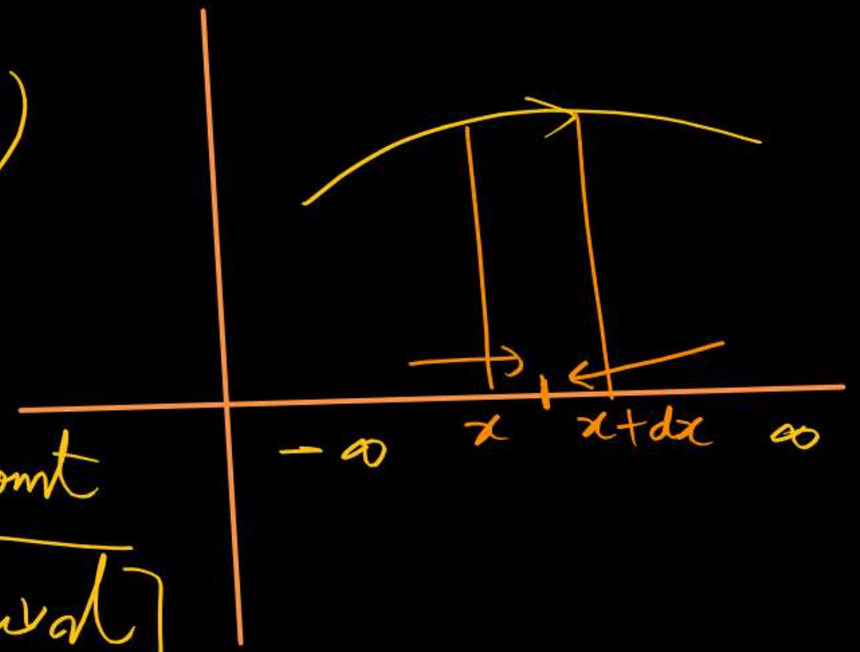
✓ Interval also given

2) $\int_{-\infty}^{\infty} f(x)\,dx = 1$ (For continuous Random)

3)
$$P[X \ge a] = \int_a^{\infty} f(x)\,dx$$
$$P[X < a] = \int_{-\infty}^{a} f(x)\,dx$$
$$P[a \le X \le b] = \int_a^b f(x)\,dx$$
$$F_X(b) - F_X(a)$$

Prob at a point

(Small interval)



Discrete

$$\sum_{i=0}^{n} P_i = 1$$

X	x_1	x_2	x_3	x_4	x_n
$P(X=x)$	P_1	P_2	P_3	P_4	P_n

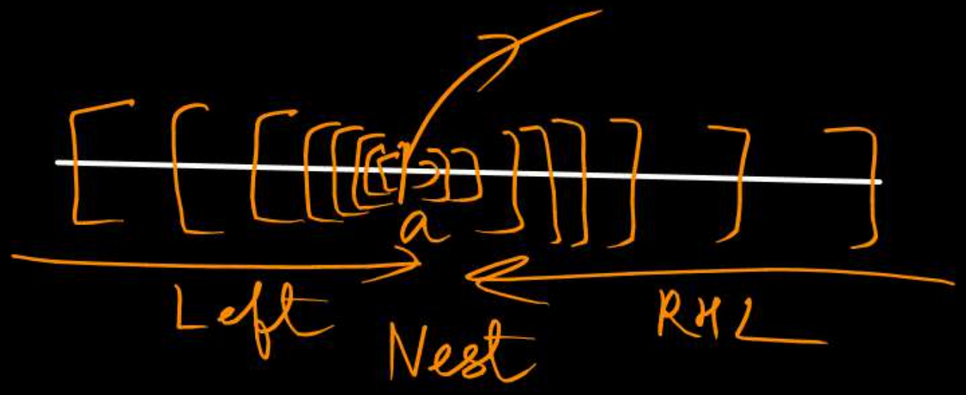
X = discrete

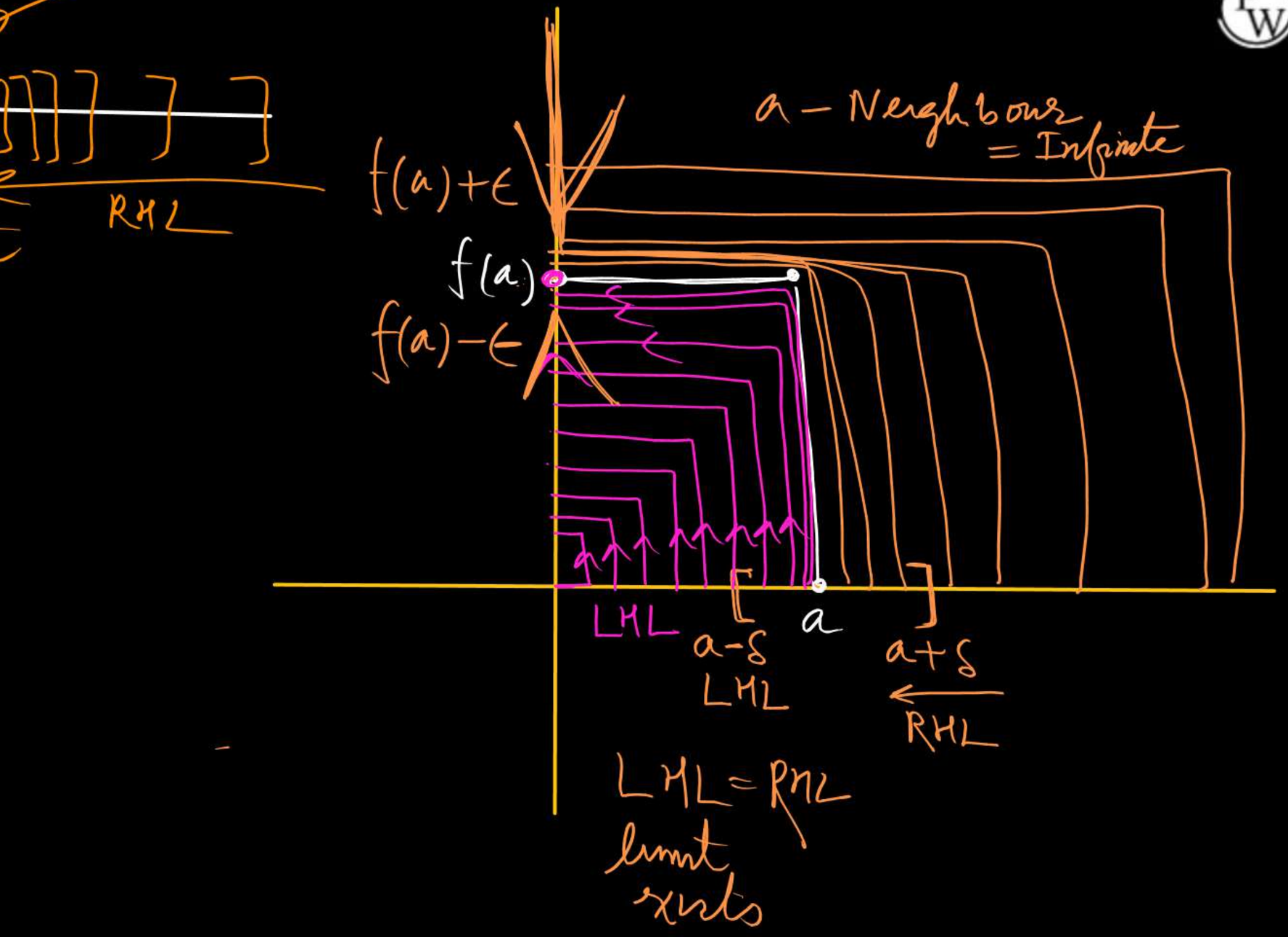
Continuous

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$P(a \le x \le b) = \int_a^b f(x)\,dx$$

X = conti


a
Left
Nest
RHL


a - Neighbours = Infinite
f(a)+ϵ
f(a)
f(a)-ϵ
LHL
a-δ
LHL
a
a+δ
RHL
LHL = RHL
limit exists



Topic :Probability and Statistics

#Q. If X is a continuous random variable whose probability density function is given by- $f(x) = \begin{cases} k(5x - 2x^2) & ,0 \leq x \leq 2 \\ 0 & ,\text{Otherwise} \end{cases}$

pdf
prob. Density function

Then $p\ (x \geq 1)$ is.

A 3/14

B 4/5

C 14/17

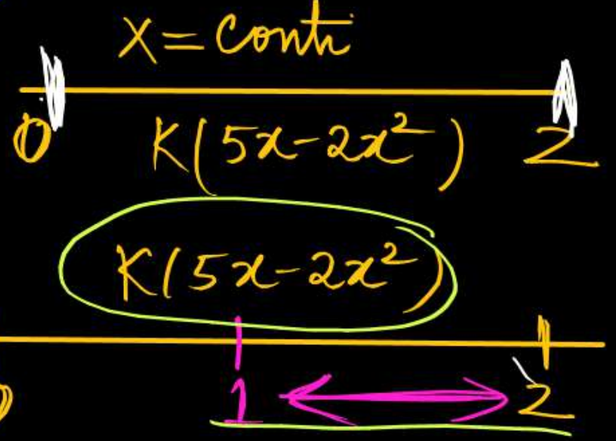
D 17/28

$0 < P(E) < 1$

$$f(x) = \begin{cases} K(5x-2x^2) & 0 \le x \le 2 \\ 0 & \text{otherwise} \end{cases}$$

$P(x \ge 1)$

$P(x \ge 1) \Rightarrow \int_1^2 K(5x-2x^2)\,dx =$ form of K



my strategy $\longrightarrow$ remove the constant

$$\int_0^2 K(5x-2x^2)\,dx = 1$$

$$K\left[\frac{5x^2}{2} - \frac{2x^3}{3}\right]_0^2 = 1$$

$$K\left[10 - \frac{16}{3}\right] = 1 \qquad K = \frac{3}{14}$$

$$\int_0^2 K(5x-2x^2)\,dx = 1$$

$$\int_a^b f(x)\,dx = 1$$

$$P(X \geq 1) = \int_1^2 \frac{3}{14}(5x - 2x^2)\,dx$$

$$= \frac{3}{14}\left[\frac{5x^2}{2} - \frac{2x^3}{3}\right]_1^2$$

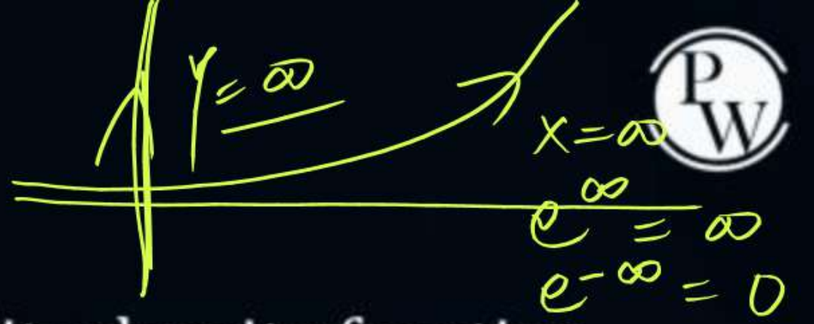
$$= \frac{3}{14}\left[\left(10 - \frac{2\times(2)^3}{3}\right) - \left(\frac{5}{2} - \frac{2}{3}\right)\right]$$

$$\boxed{P(X \geq 1) = \frac{17}{28}}$$



Topic : Probability and Statistics

$\int e^x = e^x$



#Q. A continuous random variable X has a probability density function $f(x) = e^{-x}$, $0 < x < \infty$. Then $P\{X \geq 1\}$ is-

$f(x) = \begin{cases} e^{-x} & 0 < x < \infty \\ 0 & \text{otherwise} \end{cases}$

$P(X \geq 1) = \int_1^{\infty} e^{-x}\,dx$

$\Rightarrow \left[\frac{e^{-x}}{-1}\right]_1^{\infty}$

$\Rightarrow \left[-e^{-x}\right]_1^{\infty}$

$\Rightarrow -e^{-\infty} + e^{-1}$

$= 0 + \frac{1}{e} = \frac{1}{e} = \underline{0.368}$

$P(X \geq 1)$



Interval

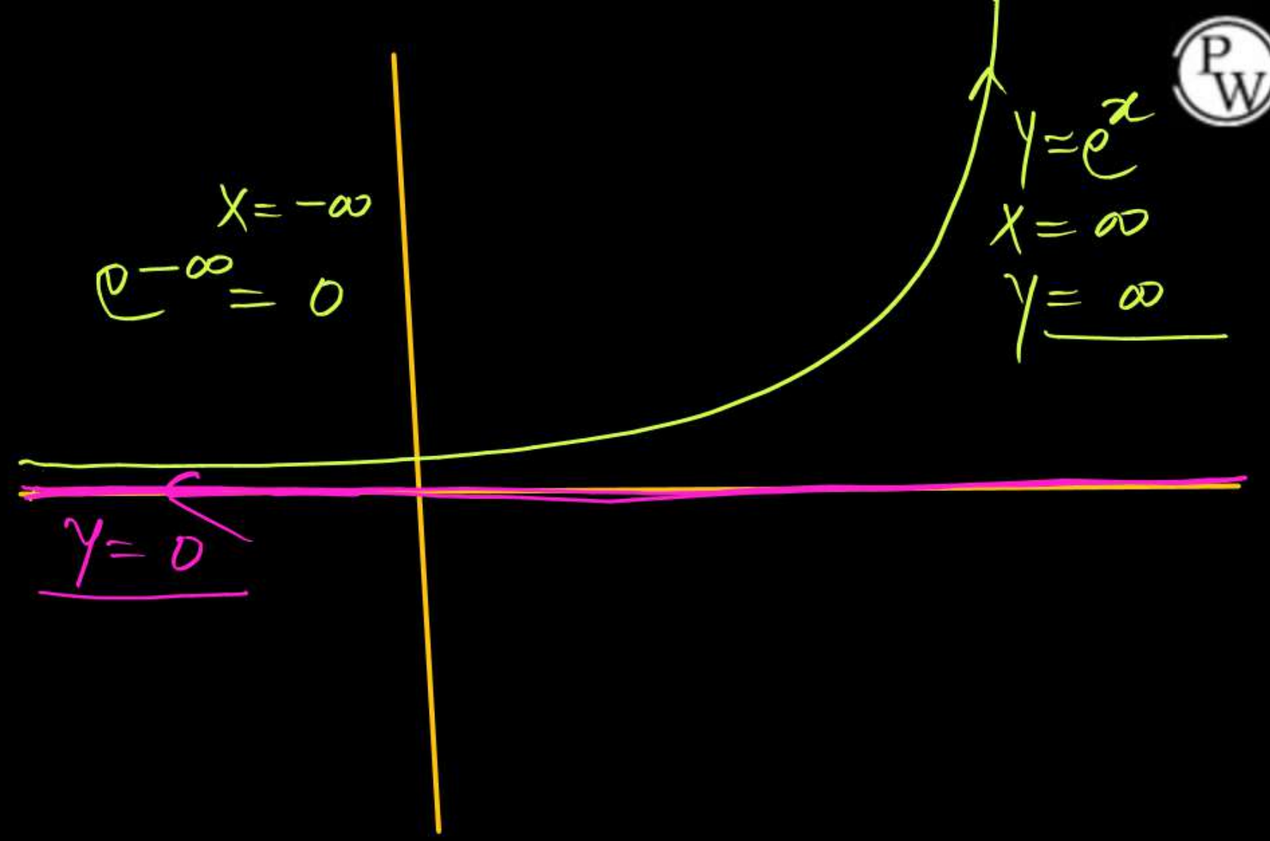
$P(X \geq a) = \int_a^{\infty} f(x)\,dx$

A 0.368

B 0.5

C 0.632

D 1.0


$y = e^x$
$x = \infty$
$y = \infty$
$x = -\infty$
$e^{-\infty} = 0$
$y = 0$



Topic :Probability and Statistics

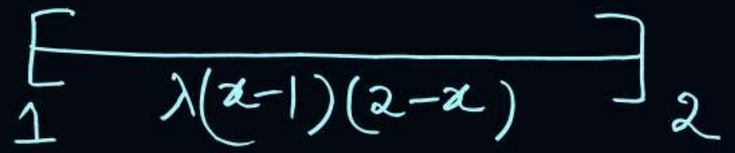
$\lambda = 6$

GATE

#Q. Find the value of λ such that the function f(x) is a valid proability density function_____.

$f(x) = \lambda\,(x-1)\,(2-x)$ for $1 \le x \le 2$

$=0$ otherwise



If Finding The constant always apply

given $\int_a^b f(x)\,dx = 1$

Interval Not given $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\int_1^2 \lambda(x-1)(2-x)\,dx = 1$$

$$\Rightarrow \lambda \int_1^2 (x-1)(2-x)\,dx = 1$$

$$\Rightarrow \lambda \left[\int_1^2 2x - x^2 - 2 + x\right] dx = 1$$

$$\Rightarrow \lambda \left[\int_1^2 (3x - x^2 - 2)\,dx = 1\right.$$

$$= \lambda \left[\frac{3x^2}{2} - \frac{x^3}{3} - 2x\right]_1^2 = 1$$

$$\frac{8-2}{6}$$

$$\lambda\left[\left(6 - \frac{8}{3} - 4\right) - \left(\frac{3}{2} - \frac{1}{3} - 2\right)\right] = 1$$

$$= \lambda\left[\left(2 - \frac{8}{3}\right) - \left(\frac{7}{6} - 2\right)\right] = 1$$

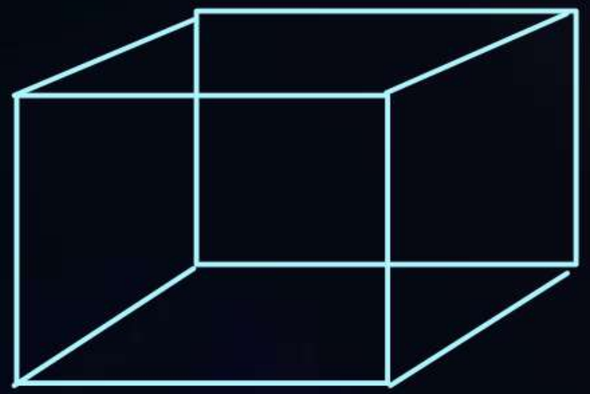
$$= \boxed{\lambda = 6}$$



Topic : Probability and Statistics

$\frac{1}{7}$ ✓

#Q. Consider a die with the property that the probability of a face with 'n' dots showing up is proportional to 'n'. The probability of the face with three dots showing up is _____.



Die
Single.



(Arrival)
(Discrete Random)

Dot $\propto$ Prob(Dot)

$1 \propto 1$	$1 = k \cdot 1$
$2 \propto 2$	$2 = 2 \cdot k$
$3 \propto 3$	$3 = 3 \cdot k$
$4 \propto 4$	$4 = 4 \cdot k$
$5 \propto 5$	$5 = 5 \cdot k$
$6 \propto 6$	$6 = 6 \cdot k$

$X = 1\ 2\ 3\ 4\ 5\ 6$

Dots	1	2	3	4	5	6	
Prob	k	2k	3k	4k	5k	6k	

Prob(Three dots) = 3k

$$K+2K+3K+4K+5K+6K=1$$

$$\Rightarrow \quad 21K = 1$$

$$\Rightarrow \quad K = \frac{1}{21}$$

$$P(3 \text{ dots}) = 3K = 3 \times \frac{1}{21} = \frac{1}{7}$$



Topic :Probability and Statistics

#Q. Lifetime of an electric bulb is a random variable with dentsity $f(x) = kx^2$ where x is measured in years. If the minimum and maximum lifetimes of bulb are 1 and 2 years repsectively, then the value of k is____.

$f(x) = kx^2$

$1y \le X \le 2$ YEARS

value of k

$f(x) = kx^2$

$$\int_1^2 kx^2\,dx = 1$$

$$= k\left[\frac{x^3}{3}\right]_1^2 = k\left[\frac{8}{3} - \frac{1}{3}\right] = 1$$

$$k = \frac{3}{7}$$



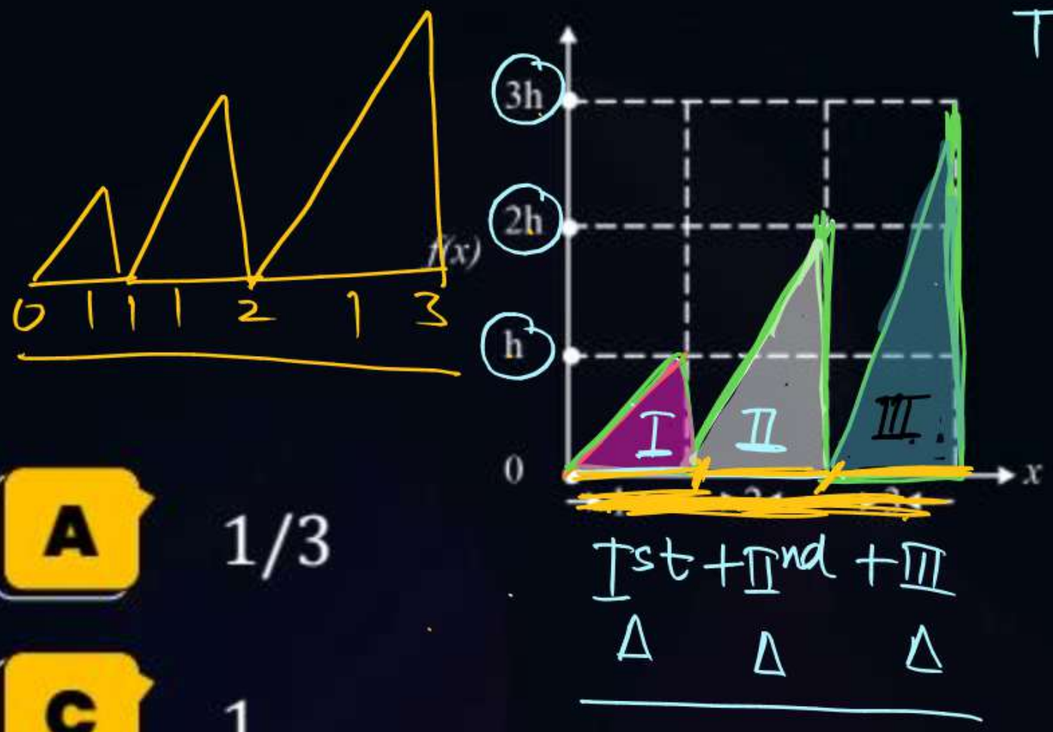
Topic : Probability and Statistics

#Q. Given that is x is a random variable in the range $[0, \infty]$ with a probability density function $\frac{e^{-\frac{x}{2}}}{k}$ the value of the constant K is.

M.W

$K=2$

#Q. The graph of function $f(x)$ is shown in figure for $f(x)$ to be valid probability density function, the value of h is-



Total Area = 1

$f(x)$ is a valid prob Density Function

$$\int_a^b f(x)\,dx = 1$$

Area of $\Delta = \frac{1}{2} \times b_1 \times h_1 + \frac{1}{2} b_2 \times h_2 + \frac{1}{2} b_3 h_3 = 1$

I st + II nd + III

$\Delta \quad \Delta \quad \Delta$

$$\frac{h}{2} + \frac{2h \times 1}{2} + \frac{3h \times 1}{2} = 1$$

$$\frac{h}{2} + h + \frac{3h}{2} = 1$$

$h = 1/3$

A 1/3

B 2/3

C 1

D 3



Topic : Probability and Statistics

#Q. $P_x(X) = Me^{(-2|x|)} + Ne^{(-3|x|)}$ is the probability density function for the real random variable X, over the entire x-axis, M and N are both positive real numbers. The equation relating M and N is-

A $M + 2/3\ N = 1$

B $2M + 1/3\ N = 1$

C $M + N = 1$

D $M + N = 3$



Topic :Probability and statistics

#Q. A normal random variable X has the following probability density function

$$fx(x) = \frac{1}{\sqrt{8\pi}} e^{-\left\{\frac{(x-1)^2}{8}\right\}}, -\infty < x < \infty$$

Then $\int_1^{\infty} f_x(x)\,dx =$

A 0

B 1/2

C $1 - \frac{1}{e}$

D 1

THANK - YOU